

DATSECURE ENTERPRISE NETWORK SECURITY IMPLEMENTATION PROJECT

Project Report

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Project Title:

DATSECURE Enterprise Network Security Implementation Using Cisco Packet Tracer

Project Type:

Network Design, Routing, Access Control, and Server Security Implementation

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1. Executive Summary

The DATSECURE Enterprise Network Security Implementation Project was developed using Cisco Packet Tracer to simulate a secure departmental network infrastructure. The project demonstrates how network segmentation, routing, and access control mechanisms can be used to secure organizational resources while maintaining necessary communication between departments.

The network consists of three departments:

- Administration
- Sales
- Human Resources (HR)

Each department operates on a separate subnet connected through a centralized router. Access Control Lists (ACLs) were implemented to enforce organizational security policies and restrict unauthorized communication between departments.

An internal web server was also deployed within the administrative network to simulate a shared business resource accessible only to approved departments.

This project demonstrates practical implementation of enterprise networking concepts including subnetting, routing, ACL security, and server access management.

2. Project Overview

Business Requirements

The organization required a network capable of:

- Supporting multiple departments.
- Separating departmental traffic.
- Providing secure inter-department communication.
- Restricting access to sensitive resources.
- Hosting internal web services.
- Enforcing centralized security policies.

Project Objectives

The primary objectives were:

- Design a scalable network architecture.
- Configure inter-network communication.
- Implement subnetting and IP addressing.
- Enforce security using ACLs.
- Deploy an internal web server.
- Restrict server access based on departmental policies.
- Validate functionality through testing and verification.

3. Network Topology and Design Criteria

Physical Network Topology

The network follows a hierarchical design consisting of:

Core Layer

The Core Layer is represented by the DATSECURE Router which performs:

- Inter-network routing
- Access Control List enforcement
- Default gateway services
- Traffic management

Access Layer

Three Cisco 2960 switches provide connectivity for:

- Administration Department
- Sales Department
- Human Resources Department

End Devices

The network contains:

- 2 Administrative PCs
- 2 Sales PCs
- 2 HR PCs
- 1 Internal Web Server

Network Architecture

The router connects directly to each departmental switch.

The architecture ensures:

- Network segmentation
- Centralized management
- Security policy enforcement
- Easy scalability

4. Network Components

Device Type	Quantity	Purpose
Cisco Router	1	Routing and ACL Enforcement
Cisco 2960 Switch	3	Department Connectivity
PCs	6	End User Workstations
Server	1	Internal Web Services

5. IP Addressing and Subnetting Plan

Network Addressing Scheme

Department	Network Address	Subnet Mask	Default Gateway
Admin	192.168.10.0/24	255.255.255.0	192.168.10.1
Sales	192.168.20.0/24	255.255.255.0	192.168.20.1
HR	192.168.30.0/24	255.255.255.0	192.168.30.1

Host Address Allocation

Administrative Department

Device	IP Address
Router Interface	192.168.10.1
Admin-PC1	192.168.10.2
Admin-PC2	192.168.10.3
DATSECURE Server	192.168.10.10

Sales Department

Device	IP Address
Router Interface	192.168.20.1
Sales-PC1	192.168.20.2
Sales-PC2	192.168.20.3

Human Resources Department

Device	IP Address
Router Interface	192.168.30.1
HR-PC1	192.168.30.2
HR-PC2	192.168.30.3

6. Device Configuration Documentation

Router Configuration

The router serves as the central routing device.

Interface Assignments

Interface	IP Address	Purpose
GigabitEthernet0/0	192.168.10.1	Admin Gateway
GigabitEthernet0/1	192.168.20.1	Sales Gateway
GigabitEthernet0/2	192.168.30.1	HR Gateway

Router Functions

- Routes packets between departments.
- Acts as default gateway.
- Enforces ACL security policies.
- Controls server access.

Switch Configuration

Each switch serves one department.

Functions include:

- MAC address learning.
- Frame forwarding.
- Broadcast containment.
- User connectivity.

Cisco 2960 switches use Spanning Tree Protocol (STP) by default to prevent Layer-2 loops.

7. Routing and Switching Protocols

Routing Method

Directly Connected Routing

The router uses directly connected routes for all three networks.

Because each subnet is directly attached to the router, no dynamic routing protocol is required.

Connected Routes

192.168.10.0/24 → Admin Network

192.168.20.0/24 → Sales Network

192.168.30.0/24 → HR Network

Switching Operations

The switches operate at Layer 2 and perform:

- MAC address table learning.
- Frame forwarding.
- Traffic segmentation.

Spanning Tree Protocol

STP provides:

- Loop prevention.
- Network stability.
- Redundancy support.

8. Security and Access Control

Security Objectives

The network was designed to:

- Protect internal resources.
- Restrict unauthorized access.
- Enforce departmental policies.
- Secure server communications.

Access Control Policy

Administration Department

Permissions:

- Access Sales Network.
- Access HR Network.
- Access Internal Server.

Access Level:

FULL ACCESS

Sales Department

Permissions:

- Access Admin Network.
- Access Internal Server.

Restrictions:

- Cannot Access HR Network.

Access Level:

LIMITED ACCESS

Human Resources Department

Restrictions:

- Cannot Access Admin Network.
- Cannot Access Sales Network.
- Cannot Access Internal Server.

Access Level:

HIGHLY RESTRICTED

Access Control List Implementation

ACLs were configured on the router to enforce organizational security policies.

The ACL rules:

- Permit Admin traffic.
- Permit Sales access to approved resources.
- Deny HR access to internal resources.
- Control HTTP access to the internal server.

9. Server Implementation

DATSECURE Web Server

Server Specifications

Parameter Value

IP Address 192.168.10.10

Network Admin

Service HTTP

Purpose

The server hosts internal web resources for authorized users.

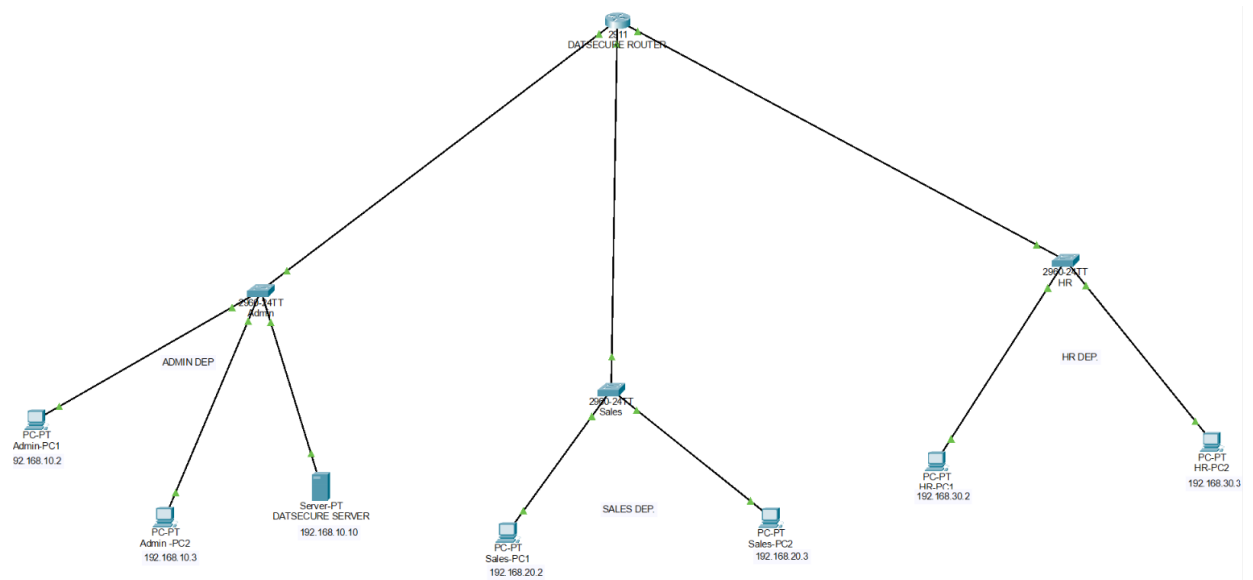
Access Rules

Department HTTP Access

Admin Allowed

Sales Allowed

HR Denied



10. Testing and Verification Plan

Phase 1 – Basic Connectivity Testing

Test 1

Source:

Admin-PC1

Destination:

Admin-PC2

Expected Result:

Successful Ping

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=10ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
  
```

Test 2

Source:

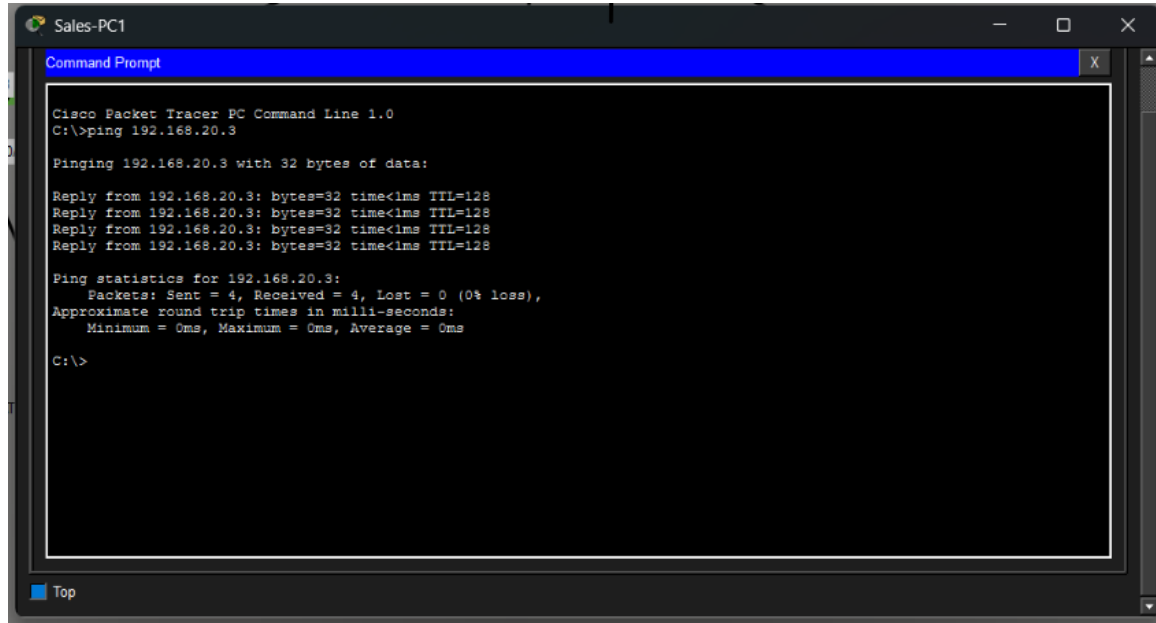
Sales-PC1

Destination:

Sales-PC2

Expected Result:

Successful Ping



```
Sales-PC1
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Test 3

Source:

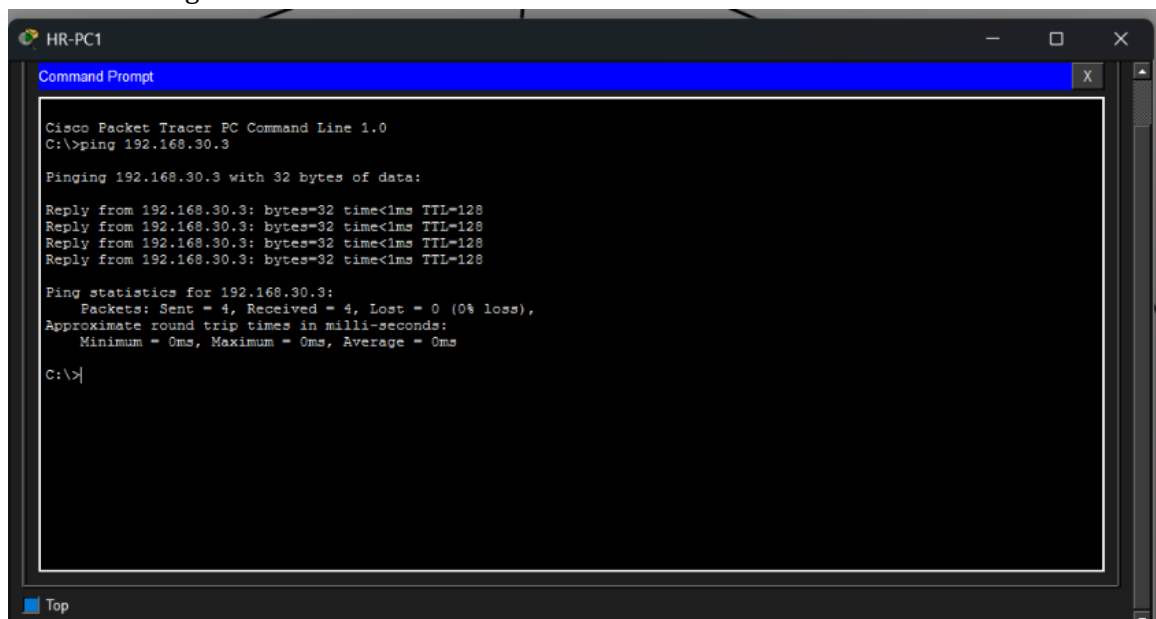
HR-PC1

Destination:

HR-PC2

Expected Result:

Successful Ping



```
HR-PC1
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

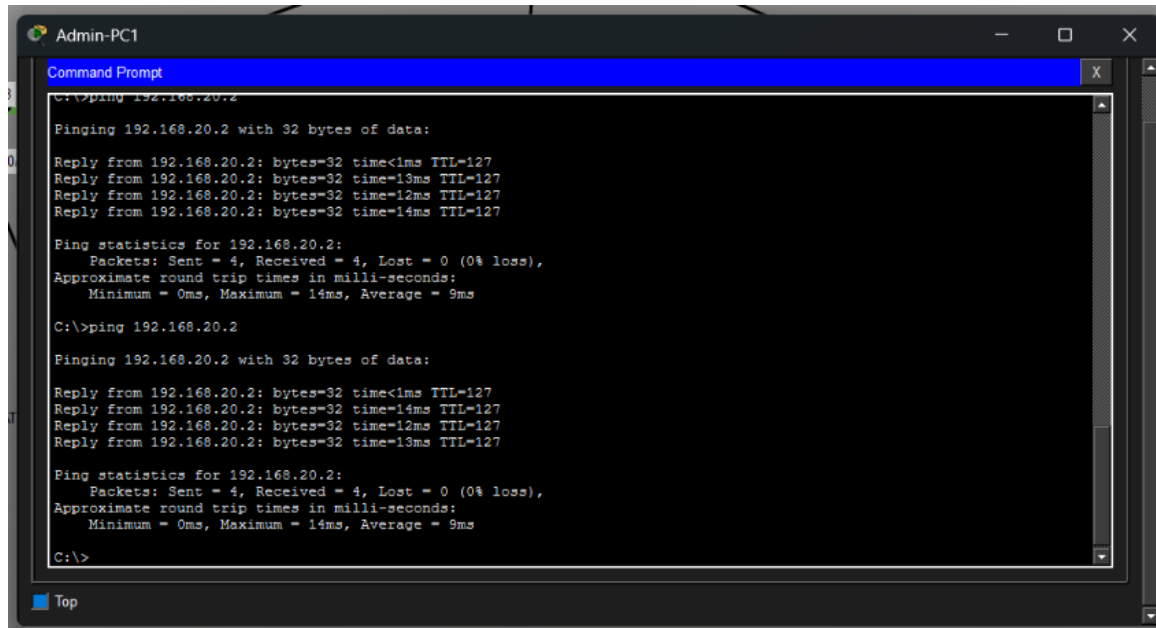
Phase 2 – Inter-Department Connectivity

Test 4

Admin-PC1 → Sales-PC1

Expected Result:

Success



```
Admin-PC1
Command Prompt
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=13ms TTL=127
Reply from 192.168.20.2: bytes=32 time=12ms TTL=127
Reply from 192.168.20.2: bytes=32 time=14ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 9ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=14ms TTL=127
Reply from 192.168.20.2: bytes=32 time=12ms TTL=127
Reply from 192.168.20.2: bytes=32 time=13ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 9ms

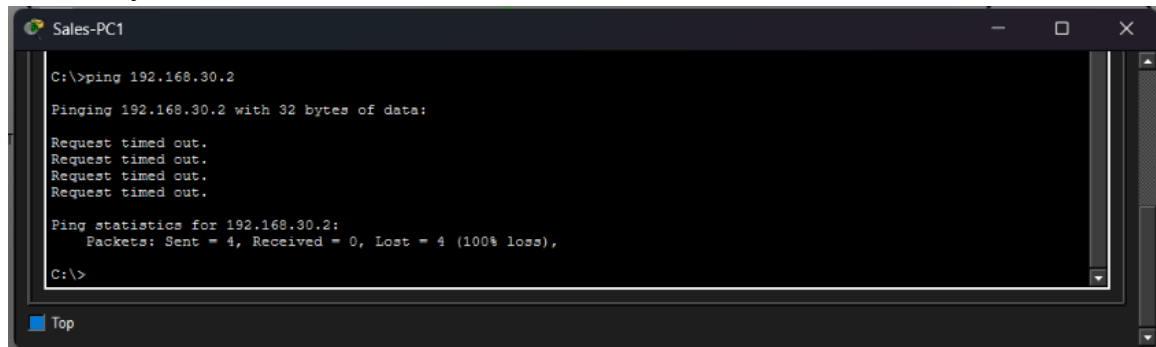
C:\>
```

Test 5

Sales-PC1 → HR-PC1

Expected Result:

Blocked by ACL



```
Sales-PC1
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

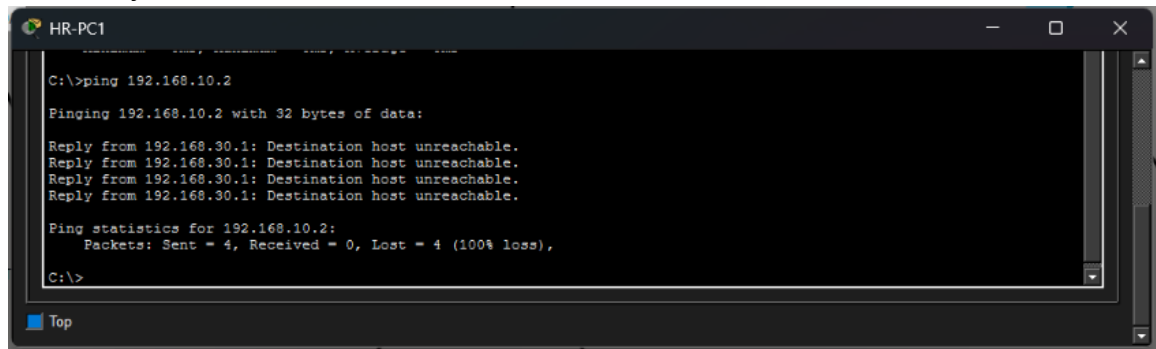
C:\>
```

Test 6

HR-PC1 → Admin-PC1

Expected Result:

Blocked by ACL



```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

Phase 3 – Server Access Testing

Test 7

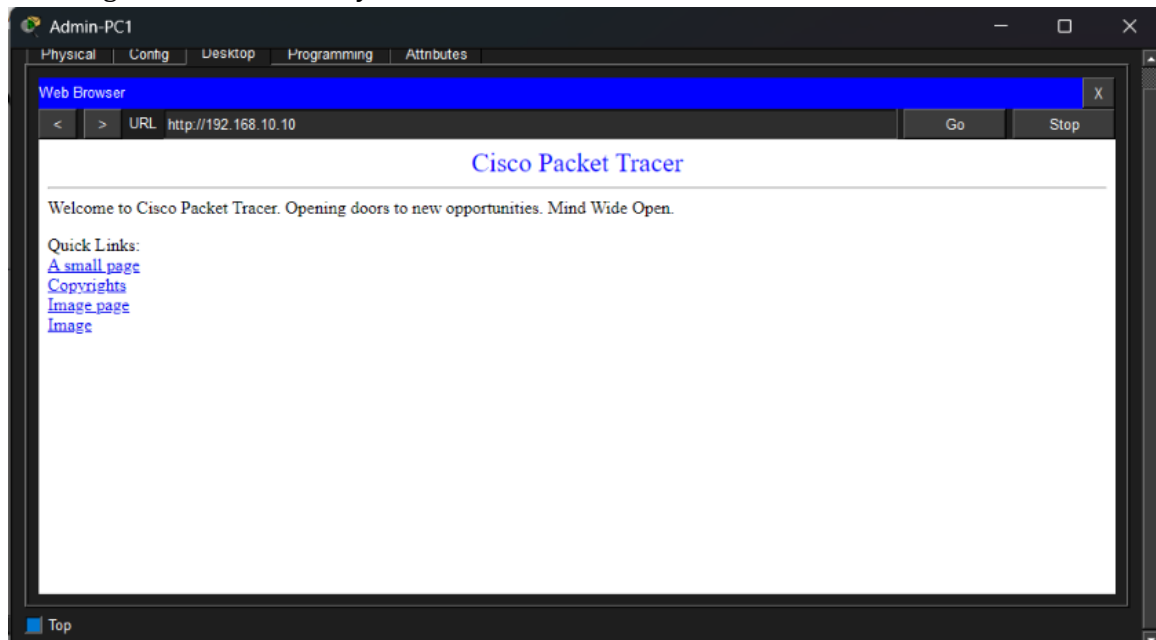
Admin-PC1

Browse:

<http://192.168.10.10>

Expected Result:

Web Page Loads Successfully



Test 8

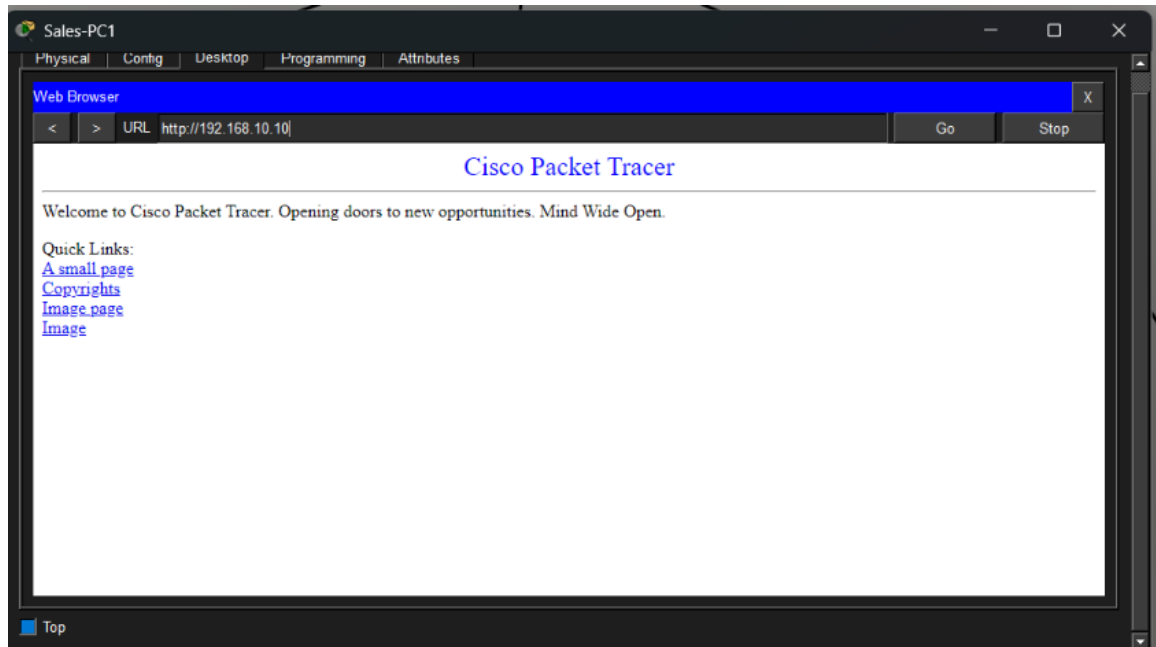
Sales-PC1

Browse:

<http://192.168.10.10>

Expected Result:

Web Page Loads Successfully



Test 9

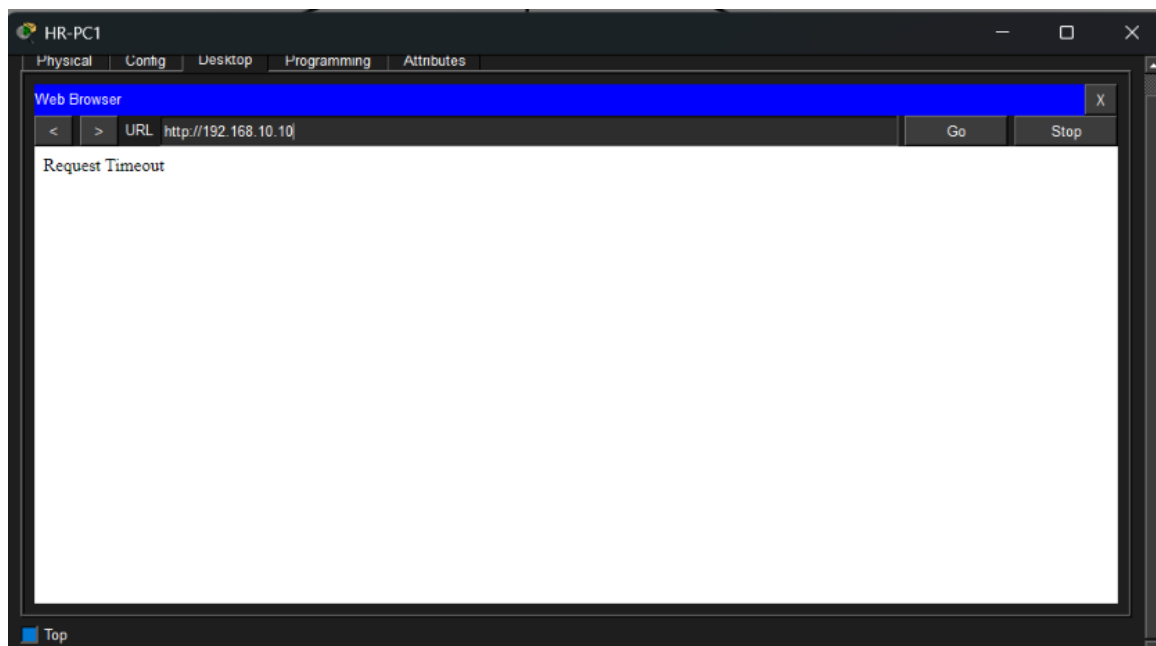
HR-PC1

Browse:

<http://192.168.10.10>

Expected Result:

Access Denied



11. Results and Analysis

All network objectives were successfully achieved.

Successful Outcomes

- ✓ Inter-department routing operational
- ✓ ACL enforcement operational
- ✓ Server accessible to authorized departments
- ✓ Unauthorized access successfully blocked
- ✓ Network segmentation functioning correctly
- ✓ End-to-end connectivity verified

12. Conclusion

The DATSECURE Enterprise Network Security Implementation Project successfully demonstrates the deployment of a secure and scalable enterprise network using Cisco Packet Tracer. Through network segmentation, IP subnetting, centralized routing, ACL-based security policies, and controlled server access, the solution meets all business and technical requirements.

The project provides a strong foundation for future enterprise network deployments and showcases practical networking skills relevant to modern organizational environments. It highlights the importance of structured network design, security policy enforcement, and effective resource management in maintaining a secure and reliable IT infrastructure.